ENJOY AI 2025: Cyber City

Competition Rule

1. Competition theme

Cyber City is where you, a brave young space explorer, come to show your talents. You're suited up in the latest space equipment, ready to take off on an exciting journey across the stars. The city has the best upgrade stations, so you can power up your spaceship in no time, getting ready to face any space challenge that comes your way. Out on the edge of the city, a giant spaceport is always buzzing with life. Spaceships of all shapes take off and land, carrying explorers like you deeper into the vast universe. Above, giant space elevators connect planets to space, transporting supplies and people deep into the cosmos. This is a place where futuristic technology creates all amazing possibilities.

2. Competition field and environment

2.1 Competition field

The competition field is covered by a 216 × 120 cm map (see Fig. 1). The map is made from PU fabric or printed cloth, with some 2.5cm-wide black guide lines on it. The robot base is in the lower-left corner of the map, and is 30 × 30 cm in size. The map must lay on a flat surface; otherwise, the performance of your robot may be affected.

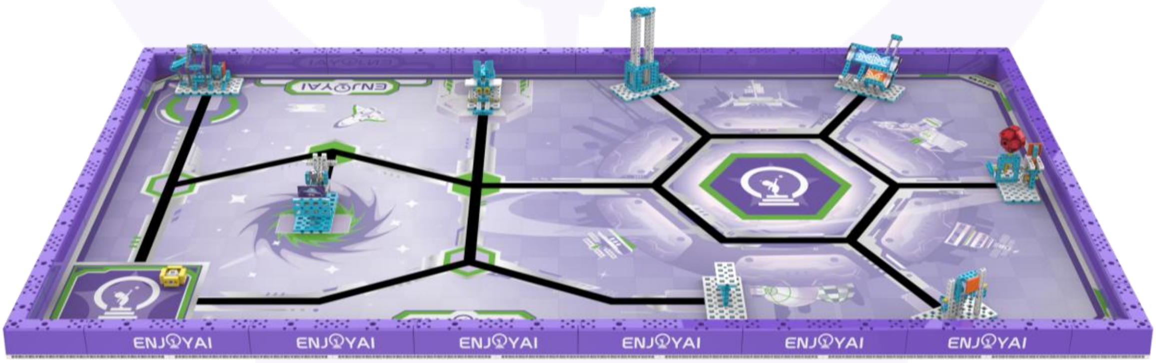

Fig. 1 Competition field

2.2 Competition environment

The competition field has fixed edges. The field should be lighted with cool low-brightness light, and should be free from magnetic interference. The actual competition environment can be affected by factors like field surface textures, uneven ground, lighting, and how task models are set up. We suggest that you take all these factors into account when designing your robots.

3. Tasks and scoring

The tasks are set in fictional scenarios. Do not confuse them with real-life situations.

3.1 Space portal

3.1.1 A space portal model is set up on the field, with a spaceship on its left and an orange panel on its right (see Fig. 2).

3.1.2 Scoring rule: If the orange panel lies flat and the magnets attract each other, the spaceship passes through the space portal (see Fig. 3). In this case, you get 50 points.

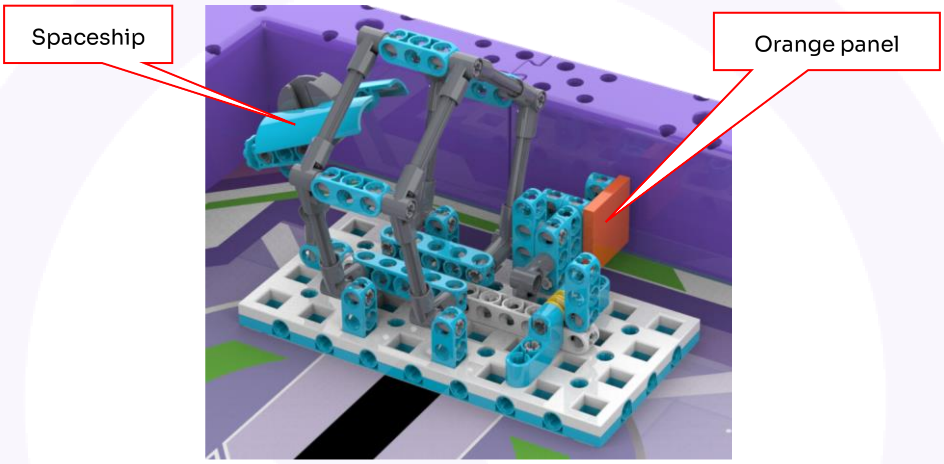

Fig. 2 Initial state

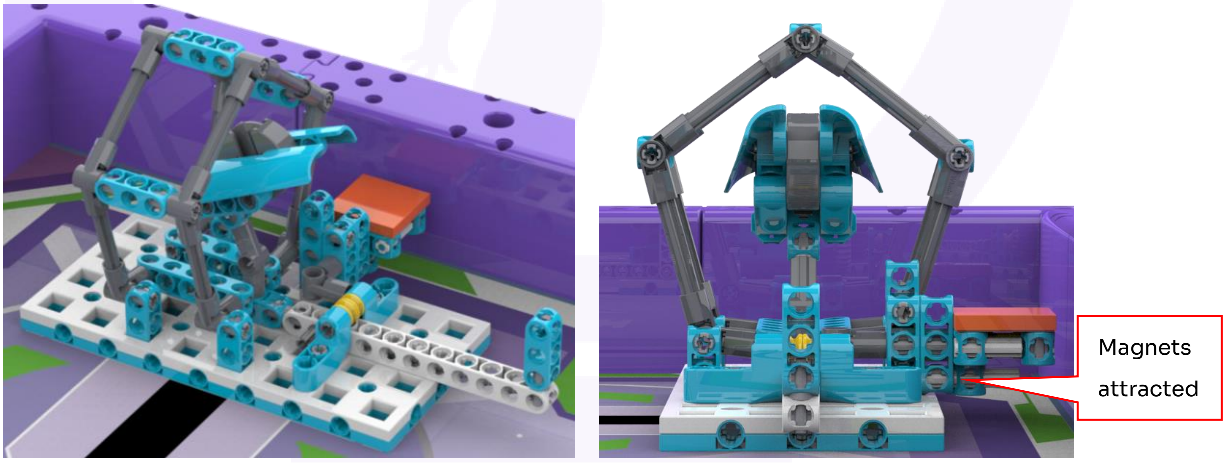

Fig. 3 Final state

3.2 Time travel

3.2.1 A time travel machine is set up on the field, with the red pointer pointing down and the lever staying up high (see Fig. 4).

3.2.2 Scoring rule: If the red pointer stays horizontally higher than the yellow axle sleeve, the time travel machine starts successfully (see Fig. 5). In this case, you get 50 points.

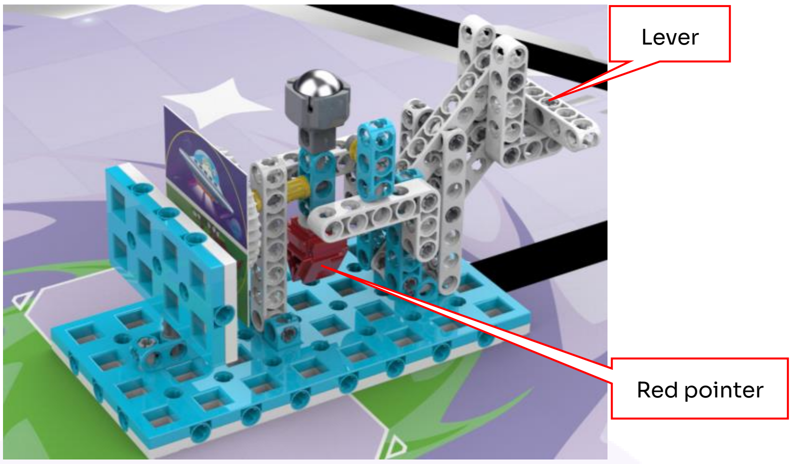

Fig. 4 Initial state

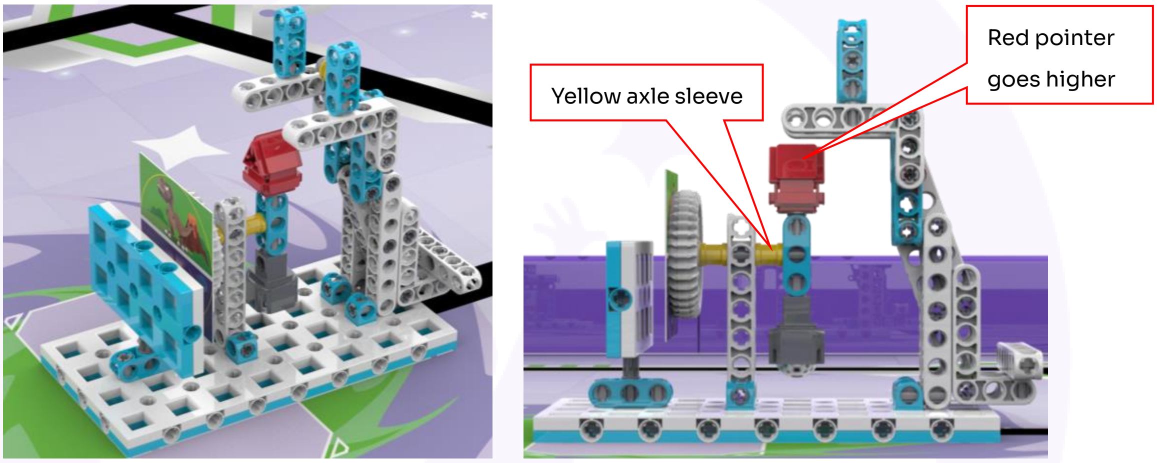

Fig. 5 Final state

3.3 Spaceship

3.3.1 A spaceship is set up on the field, with the toggle pulled completely out and two wings folded (see Fig. 6).

3.3.2 Scoring rule: If the two wings stay horizontally lower than the 20 beam on the spaceship's head, the wings successfully unfold (Fig. 7). In this case, you get 40 points.

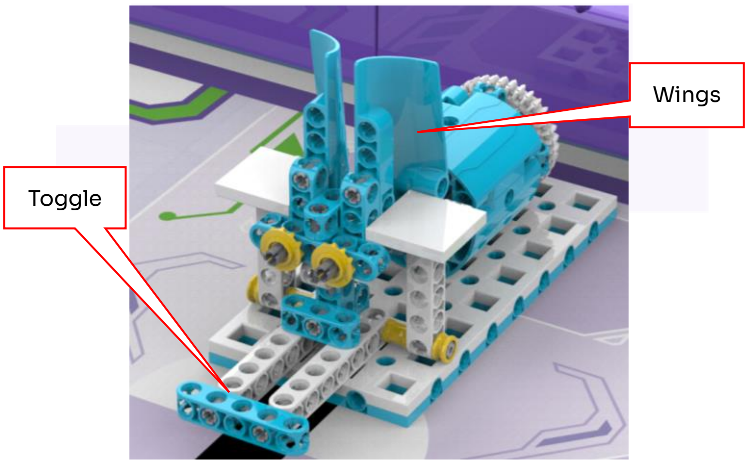

Fig. 6 Initial state

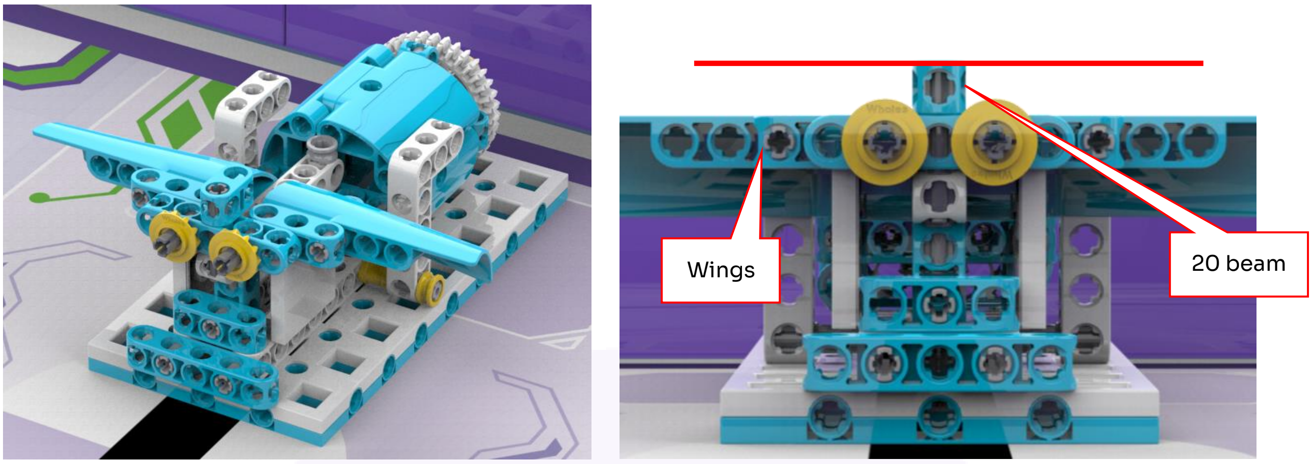

Fig. 7 Final state

3.4 Plant growing

3.4.1 A plant cultivation center is set up on the field, and you'll need to transport some plants into it (see Fig. 8).

3.4.2 Scoring rule: When the plants are put onto the white platform and do not touch the field map, you get 60 points (see Fig. 9).

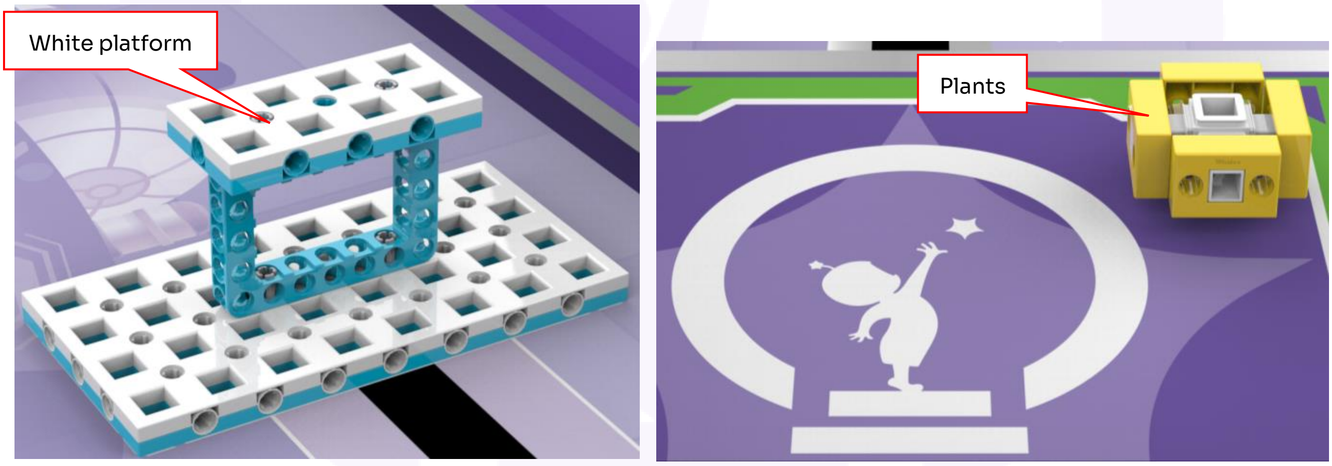

Fig. 8 Initial state

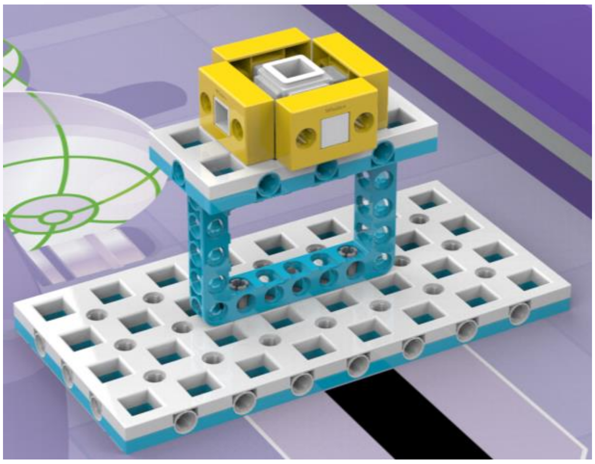

Fig. 9 Final state

3.5 Space elevator

3.5.1 A space elevator is set up on the field, and the elevator cab lies on the first floor (see Fig. 10).

3.5.2 Scoring rule: If the elevator cab's magnet attracts the track's magnet, the cab rests on the top floor (see Fig. 11). In this case, you get 50 points.

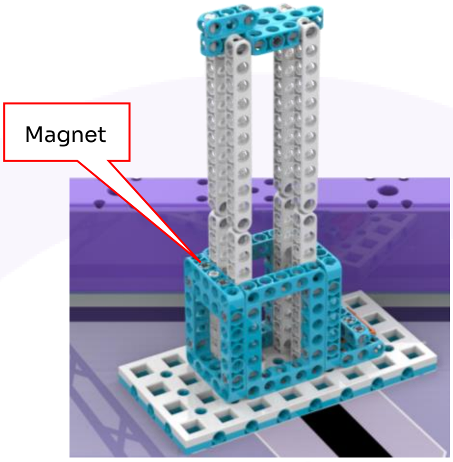

Fig.10 Initial state

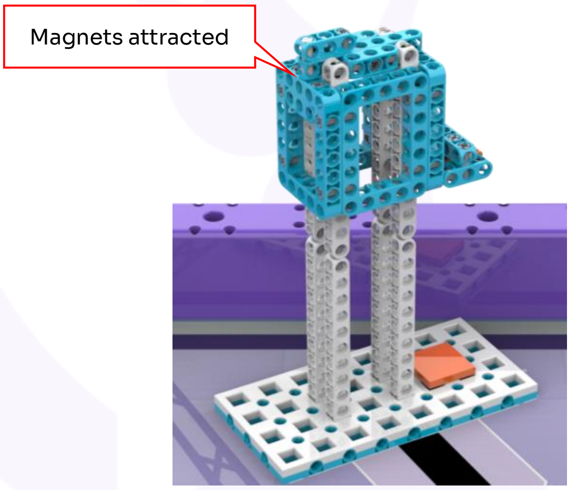

Fig. 11 Final state

3.6 Genetic lab

3.6.1 A genetic lab is set up on the field, and the initial state is shown in Fig. 12.

3.6.2 Scoring rule: If the orange genome connects to the blue genome through magnets, you get 40 points (see Fig. 13).

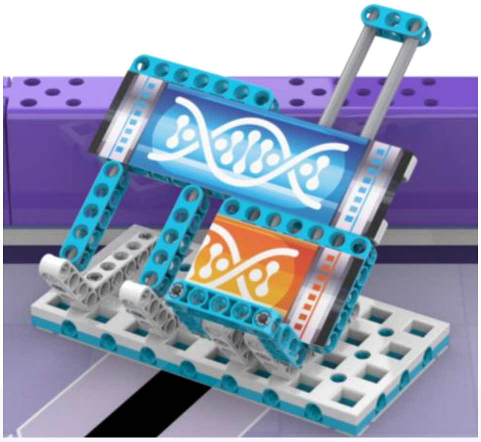

Fig. 12 Initial state

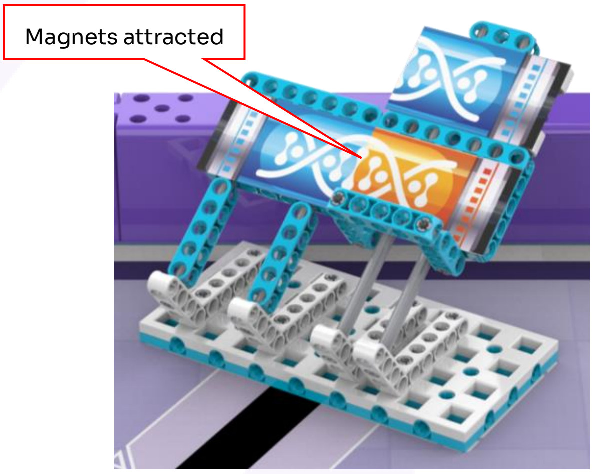

Fig. 13 Final state

3.7 Sun simulator

3.7.1 A sun simulator is set up on the field, with the orange panel folded and the lever turned at a random angle (see Fig. 14).

3.7.2 Scoring rule: If the top-down projection of the orange panel is on the right of the 90° white beam, the panel lies down successfully (see Fig. 15). In this case, you get 50 points.

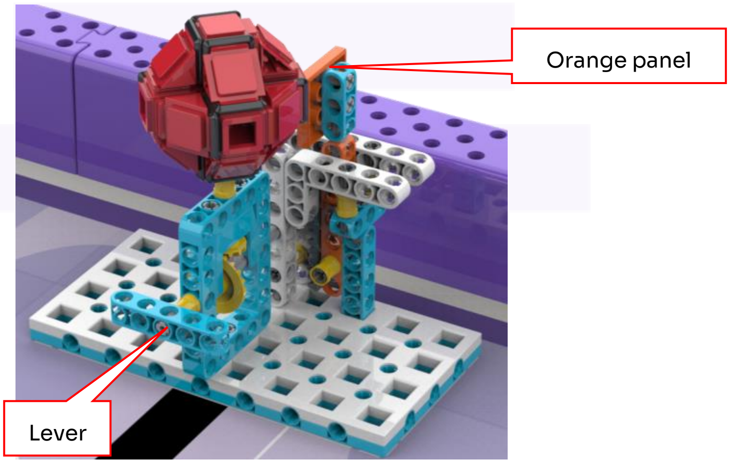

Fig. 14 Initial state

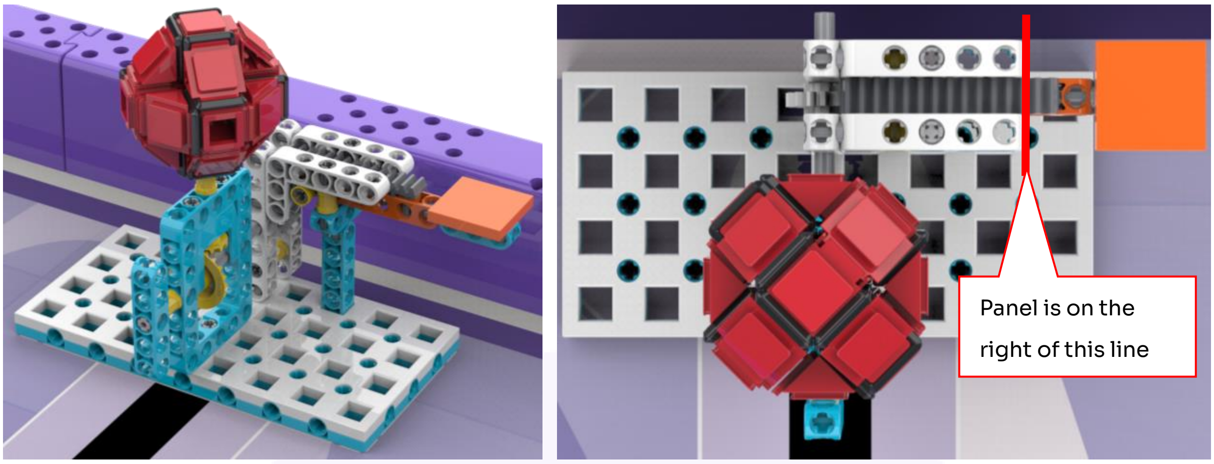

Fig. 15 Final state

3.8 Brain-machine link

3.8.1 A brain-machine link device is set up on the field, with the headset flipped back (see Fig. 16).

3.8.2 Scoring rule: If the 2 double bolt touches the 90° 2×3 simple beam, the headset is put on (see Fig. 17). In this case, you get 60 points.

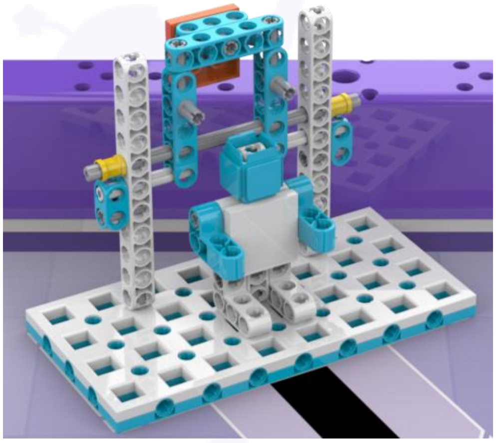

Fig. 16 Initial state

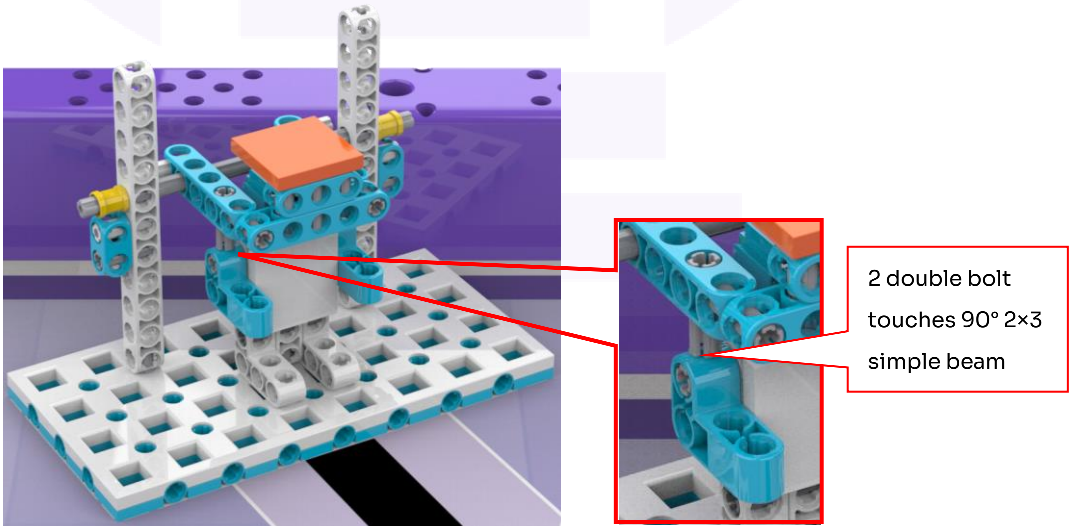

Fig. 17 Final state

3.9 Bonus task

3.9.1 There may be a bonus task in the competition. If there is one, the task model and specific rules will be revealed by the chief referee in the debugging phase.

3.9.2 If there is a bonus task, it may replace one of the tasks from 3.1 to 3.8.

3.10 Task models

The position of the model for the "plant growing" task is fixed. For the models of other tasks, the positions will be announced before the competition starts.

4. Robot

4.1 Robot size: Before the robot leaves the robot base, its size cannot exceed 30 × 30 × 30 cm. After it leaves the robot base, some of its parts can unfold.

4.2 Controller: You cannot change your controller during the competition. You can use only one controller for each robot.

4.3 Actuators: You can use at most four non-servo motors for each robot in a round. You cannot use servos for your robot.

4.4 Sensors: The number and types of sensors used for your robot are not limited.

4.5 Connecting method: The parts of your robot can only be connected using plastic snap-fit parts. You cannot use 3D-printed parts. You cannot use rubber bands, zip ties, screws, adhesives, or tapes to connect parts together.

4.6 Power supply: The robot must be powered by its original battery pack (no external power sources allowed). The battery voltage cannot exceed 9V, and you cannot use circuits to change the voltage.

4.7 Each team can use only one robot. You cannot share your robot with other teams.

5. Competition

5.1 Team

5.1.1 Each team must have two students and one coach. The students must still be in school by June, 2025.

5.1.2 Contestants should proactively deal with problems they may meet during the competition. They must respect and politely communicate with their teammates, opponents, volunteers, referees, and all those who contribute to the competition.

5.2 Competition format

5.2.1 The competition will have three divisions: elementary school, middle school, and high school.

5.2.2 There will be no separate preliminary or final rounds. Every team will have an equal number of chances to compete, and every round will be scored.

5.2.3 The tasks to complete will be announced before the competition starts. They will be selected from Task 3.1 to Task 3.8 and there may be a bonus task. The number of tasks that need to complete may be different for different divisions.

5.2.4 At the end of the whole competition, the teams are ranked by point totals of all rounds.

5.2.5 The organizing committee may change the format based on the number of signed-up teams and the actual venue setup.

5.3 Procedure

5.3.1 Build robot in advance

5.3.1.1 You can code and debug your robot only in the designated area of the competition venue.

5.3.1.2 You must check in before you go to the preparation area. You must carry with you only facilities, tools and parts required by this competition. The referees will check the things you carry. You can build your robot before you enter the preparation area.

5.3.1.3 During the competition, you cannot go online, download things, take photos of the competition field, or contact your coach or parents in any way.

5.3.1.4 You will have a set amount of time to debug your robots before the competition starts. After this time, you must put your robot in the designated area, and you cannot modify or download programs until the round ends.

5.3.1.5 After a round ends, you can fix your robots and update control programs in the preparation area. Pay attention to the time of the next round. Make sure that you attend each round in the required sequence.

5.3.2 Pre-match preparation

5.3.2.1 Before the match, take your robot with you and follow a guide into the competition field. If you and your teammate don’t show up on time, your team will be considered to have given up.

5.3.2.2 After it is your turn in the competition, go close to the robot base. Do not lean against the edges around the competition field.

5.3.2.3 Put your robot into the robot base. The top-down projection of the robot (including things carried by it for some specific tasks) cannot stick partially or entirely out of the robot base.

5.3.2.4 Then, make preparations within one minute. During the preparation, your robot cannot leave the robot base, and you cannot modify or download programs. Once you are ready, signal to your referee.

5.3.3 Start of match

5.3.3.1 After the match starts, your robot starts moving on its own.

5.3.3.2 When referees confirm that a team is ready, they will count down "3, 2, 1, start". After hearing "start", you can start your robot.

5.3.3.3 If the robot moves before the "start" command, it will be a false start, and your team will get a warning or penalty.

5.3.3.4 After the robot starts, it can only be controlled by the program in its controller.

5.3.3.5 After the robot starts, no parts should come off or fall on the field. If a part accidentally falls, the referee will take it off the field, and it cannot be used in the current round again. If you intentionally disassemble a part to complete a specific task, you do not get points for this task.

5.3.3.6 Once the match starts, if a task model is taken off the field (except those carried back by the robot as required in a task), it cannot return onto the field until the current round ends.

5.3.4 Retry

5.3.4.1 The robot needs to retry if:

(1) A team member touches the robot after it leaves the robot base.

(2) The robot goes off the competition field.

5.3.4.2 During the retry, you cannot change the layout of the field, and you must put the robot back to the robot base.

5.3.4.3 Tasks that are completed before the retry will still count. However, if you're retrying to return to the robot base, the task models carried by your robot cannot be used and must be kept by the referee until the current round ends.

5.3.4.5 The number of retries for each round is not limited. The timer won't stop or restart during retries.

5.3.5 Return to robot base

5.3.5.1 The robot can be programmed to return to the robot base as many times as needed without being considered as retries.

5.3.5.2 The return is considered successful only if the top-down projection of the robot overlaps or becomes within the base.

5.3.5.3 After the robot returns, you can touch the robot to make changes or repairs.

5.3.6 End of match

5.3.6.1 Each round lasts for 150 seconds.

5.3.6.2 If you decide to stop your play during the competition, raise your hand to signal the referee. The referee will stop the timer and end the match. If you do not give a signal, wait until the referee announces that the match is over.

5.3.6.3 After the referee announces that the round is over, power off your robot immediately. At this time, you can no longer touch or move your robot or task models on the field. If you touch or move them, your points for related tasks are considered invalid.

5.3.6.4 The referee will announce your score. If the score is miscalculated, you can ask the referee to correct it. If it is right, sign to confirm your score. If there is a dispute, you can ask the chief referee for rejudging.

5.3.6.5 After the round ends, you must restore the layout of the competition field to the initial state and take your robot to the preparation area.

6. Scoring

6.1 After each match, points will be given based on the tasks the robot completes. If you or your robot break the model of a completed task before the match ends, no points will be given to this task. For the scoring rule of each task, see Section 3 in this document.

6.2 The order in which you complete tasks does not affect your points for a single task.

6.3 If there are no retries in a match, you get an extra 40-point reward. If there is one retry, you get 30 points. 2 retries, 20 points. 3 retries, 10 points. If there are 4 or more retries, you do not get this extra reward.

7. Fouls and disqualification

7.1 If you haven't arrived 15 minutes after the match starts, your team will be disqualified from this round.

7.2 If there is a false start for the first time, your team will get a warning. If there is a second false start, your team will be disqualified from this round.

7.3 If your robot crashes into any field device and causes damage, your team will get a warning. If it happens again, your team will be disqualified from this round.

7.4 If you or your robot breaks a task model, your team will get a warning and points will not be given to this task.

7.5 If someone not within the current team interferes with the match, the team to which the person belongs will be disqualified from the match, and the current team will restart the play.

7.6 If you touch a task model outside the robot base, that model will become invalid, and the match will stop right away. Points will be given based on the tasks completed before the touching.

7.7 If you don't listen to the referee's instructions, your team will be disqualified from the current round.

7.8 If you go online, download things, or take pictures of the competition field during the competition, your team will be disqualified from this round.

7.9 If you contact your coach or parents without being allowed by the referee, your team will be disqualified from this round.

8. Ranking

8.1 Teams in each division will be ranked by total scores. If two teams are tied, the following rules will be followed in sequence:

(1) The team that spends less time across all rounds ranks higher.

(2) The team that retries fewer times across all rounds ranks higher.

(3) The team with the higher score from any single round ranks higher.

8.2 Your rankings will decide the award you may get: champion, runner-up, or third place. You will not be ranked if you get zero points or give up the competition.

Appendix:

Scoring Sheet of Cyber City					Round ___
No.		Team		Division	

Task	Description	Full points	Scored points
Space portal	The orange panel lies flat and the magnets attract each other.	50	
Time travel	The red pointer stays horizontally higher than the yellow axle sleeve.	50	
Spaceship	The two wings stay horizontally lower than the 20 beam on the spaceship's head.	40	
Plant growing	The plants are put onto the white platform and do not touch the field map.	60	
Space elevator	The elevator cab's magnet attracts the track's magnet.	50	
Genetic lab	The orange genome connects to the blue genome through magnets.	40	
Sun simulator	The top-down projection of the orange panel is on the right of the 90° white beam.	50	
Brain-machine link	The 2 double bolt touches the 90° 2×3 simple beam.	60	
Bonus task	If there is a bonus task, the task model and specific rules will be revealed in the debugging phase.	100	
Extra reward	40 - 10 × Number of retries		
Total score			
Time spent			

Score confirmation			
I confirm that the scores in this sheet are correct.			
Team members:		Referee:	
Remarks			
Chief referee:		Scorekeeper:	